higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

2 April 2024 (X-paper)
09:00–12:00

Scientific calculators and drawing instruments may be used.

This question paper consists of 6 pages and a formula sheet of 5 pages.

076Q1A2402

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimals.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Sketches must be large, neat and fully labelled
8. Use only a black or blue pen.
9. Write neatly and legibly.

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QUESTION 1

1.1 Determine the following limit:

$$\lim_{x\to 0}\left(\frac{e^x}{e^x-1}-\frac{1}{x}\right) \quad (4)$$

1.2 Determine the value(s) of x for which $f(x)$ is discontinuous if

$$f(x)=\frac{x^2-5x+4}{e^{x^2-9}-1} \quad (2)$$

[6]

QUESTION 2

2.1 Determine from first principles, the derivative of:

$$f(x)=\frac{5}{7x+2} \quad (4)$$

2.2 Determine $\frac{dy}{dx}$ in each of the following cases:
(Simplification not required)

2.2.1 $y=\frac{1}{5}\sec^{-4}(4+x^3)$ (3)

2.2.2 $y=\frac{\sqrt{2x^3+1}}{\sqrt{2x^3-1}}$ (4)

2.3 Given:

$$y=\frac{(3x+5)^3}{\sqrt{x^4-1}}$$

Calculate $\frac{dy}{dx}$ with the aid of logarithmic differentiation. (4)

2.4 Given the implicit function: $(y^2+x^2)^3=8x^2y^2$

2.4.1 Determine $\frac{dy}{dx}$ (4)

2.4.2 Determine the equation of the tangent to the graph at the point $(-1,1)$ (3)

[22]

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QUESTION 3

3.1 Given:

$$f(x) = x^3 - 2x + 3$$

3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

3.1.2 Draw up a table of x and $f(x)$, where x is ranging from $x = -2$ to $x = 2$. (2)

3.1.3 Draw a neat graph of $f(x)$ between these values to show the turning points on it. (2)

3.1.4 One root of the equation $x^3 - 2x + 3 = 0$ is close to -1.9. Use Taylor's/ Newton's method once to determine a better approximation of this root (Root correct to THREE decimal figures). (3)

3.2 Find a pair of positive numbers that have a product of 256 and minimize the sum of four times the first number and the second number. (5)

3.3 An ice cube with a 15 cm side is melting, so that its volume decreases at 9 cm^3/s.

How fast is its surface area changing? (5)

[20]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $y = \dfrac{\tan^4\sqrt{x}.\sec^2\sqrt{x}}{\sqrt{x}}$ (3)

4.1.2 $y = \cos 2x \cos(-6x)$ (3)

4.1.3 $y = \sin^2\left(\dfrac{15}{4}x\right)$ (3)

4.1.4 $y = \dfrac{x^3 + 9x^2 + 24x + 16}{x - 1}$ (5)

4.1.5 $y = \cos x \ln(\tan x)$ (4)

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$$y = \frac{x^2 + 2x + 1}{(2x - 3)^3}$$

(7)

[25]

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QUESTION 5

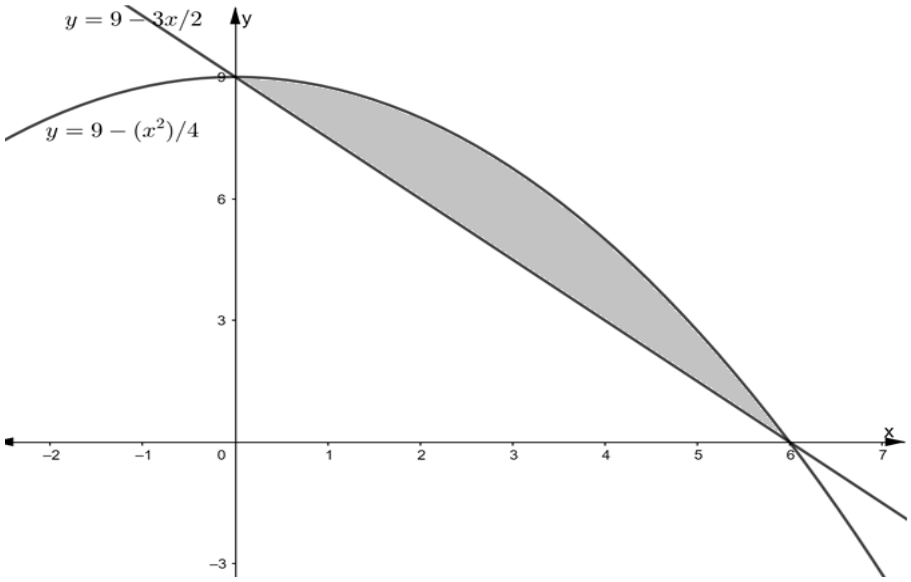
5.1 Evaluate the definite integral:

$$\int_1^5 x^4 \ln x \, dx$$

(5)

5.2 State the value of the Laplace transform: $\mathcal{L}(\frac{t}{7})$. (1)

5.3 Given below is a graph of the functions $y = 9 - \frac{3}{2}x$ and $y = 9 - \frac{x^2}{4}$, showing the area bounded by them.



5.3.1 Calculate the magnitude of the area. (3)

5.3.2 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.3.1 is rotated about the x-axis. (5)

5.4 Calculate the second moment of area of a rectangular lamina with sides 12 $cm \times 8\ cm$ and about an 8 cm side. (5)

[19]

QUESTION 6

6.1 Determine the general solution of:

$$x\frac{dy}{dx} = x + 1 + xy + y \tag{3}$$

6.2 Determine the particular solution of:

$$\frac{1}{(2-5x)^2}\frac{d^2y}{dx^2} = -3, \quad \text{if} \quad \frac{dy}{dx}_{|\ x=1} = 0 \text{ and } y(1) = -\frac{1}{4} \tag{5}$$

[8]

TOTAL: 100

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MATHEMATICS N5

FORMULA SHEET

Any applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \text{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$\sin^2 A = ½ - ½ \cos 2A$

$\cos^2 A = ½ + ½ \cos 2A$

$\sin (A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$\sin A \cos B = ½ [\sin (A + B) + \sin (A - B)]$

$\cos A \sin B = ½ [\sin (A + B) - \sin (A - B)]$

$\cos A \cos B = ½ [\cos (A + B) + \cos (A - B)]$

$\sin A \sin B = ½ [\cos (A - B) - \cos (A + B)]$

$$\tan x = \frac{\sin x}{\cos x} \; ; \; \sin x = \frac{1}{\text{cosec } x} \; ; \; \cos x = \frac{1}{\sec x}$$

BINOMIAL THEOREM

$$(x + h)^n = x^n + nx^{n-1} h + \frac{n(n-1)}{2!} x^{n-2} h^2 + \ldots$$

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DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x) \,.\, v(x)$$

$$\frac{dy}{dx} = u \,.\, \frac{dv}{dx} + v \,.\, \frac{du}{dx}$$

$$= u \cdot v' + v \cdot u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v \,.\, \frac{du}{dx} - u \,.\, \frac{dv}{dx}}{v^2}$$

$$= \frac{v \cdot u' - u \cdot v'}{v^2}$$

CHAIN RULE

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

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$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x.\ln a$	$\frac{a^x}{\ln a} + c$
$\log_e x$	$\frac{1}{x}$	—
$\log_a x$	$\frac{1}{x \ln a}$	—
$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$
$\tan x$	$\sec^2 x$	$\ln(\sec x) + c$
$\cot x$	$-\text{cosec}^2 x$	$\ln(\sin x) + c$
$\sec x$	$\sec x \cdot \tan x$	$\ln[\sec x + \tan x] + c$
$\text{cosec } x$	$-\text{cosec } x \cdot \cot x$	$\ln(\text{cosec} x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1} x$	$\frac{1}{1+x^2}$	—
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	—
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$	—
$\text{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	—

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$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
$\frac{1}{\sqrt{a^2 - x^2}}$	—	$\sin^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{a^2 + x^2}$	—	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{x\sqrt{x^2 - a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2 - x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2 - x^2} + c$
$\frac{1}{x^2 - a^2}$	—	$\frac{1}{2a} ln\left(\frac{x - a}{x + a}\right) + c$
$\frac{1}{a^2 - x^2}$	—	$\frac{1}{2a} ln\left(\frac{a + x}{a - x}\right) + c$

INTEGRATION

$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x)\, g(x)\, dx$

$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$

$\int [f(x)]^n f'(x)\, dx = \frac{[f(x)]^{n+1}}{n+1} + c$

$\frac{f(x)}{(ax + b)(cx + d)} = \frac{A}{ax + b} + \frac{B}{cx + d}$

$\frac{f(x)}{(x + a)^n} = \frac{A}{(x + a)} + \frac{B}{(x + a)^2} + \frac{C}{(x + a)^3} + \cdots \frac{Z}{(x + a)^n}$

APPLICATIONS OF INTEGRATION

AREAS

$A_x = \int_a^b y dx \; ; \; A_x = \int_a^b (y_1 - y_2) dx$

$A_y = \int_a^b x dy \; ; \; A_y = \int_a^b (x_1 - x_2) dy$

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VOLUMES

$$V_x = \pi \int_a^b y^2 dx \; ; V_x = \pi \int_a^b \left(y_1^2 - y_2^2\right) dx$$

$$V_y = \pi \int_a^b x^2 dy \; ; V_y = \pi \int_a^b \left(x_1^2 - x_2^2\right) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2 dA \; ; I_y = \int_a^b r^2 dA$$

MOMENTS OF INERTIA

Mass = density × volume

$M = \rho V$

DEFINITION: $I = m r^2$

GENERAL: $I = \int_a^b r^2 dm = \rho \int_a^b r^2 dV$

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